

Loss Functions in DL.

- Regression
 - MSE
 - MAE
 - huber loss
- Classification
 - binary cross entropy
 - categorical cross entropy
 - hinge loss
- GAN
 - discriminator loss
 - minmax gan loss
- Autoencoders
 - KL divergence
- Embedding
 - Triplet loss
- Object Detection
 - ti-focal loss

Loss Function vs cost function.

- A loss function calculates the error for a single training example
- A cost function is the average or total of the loss functions across the entire dataset.

ex.,

loss function: $(y_i - \hat{y}_i)^2$

cost function: $\frac{1}{n} \sum (y_i - \hat{y}_i)^2$

cgpa	iq	Package	prediction	loss function
6.3	100	6.3	6.1	$(6.3 - 6.1)^2$
7.1	91	4.1	4.0	$(4.1 - 4)^2$
8.5	83	3.5	3.7	$(3.5 - 3.7)^2$

cost function = $\frac{1}{3} ((6.3 - 6.1)^2 + (4.1 - 4)^2 + (3.5 - 3.7)^2)$

① Huber Loss ^{work as MSE} for $|y - \hat{y}| \leq \delta$

$$L = \begin{cases} \frac{1}{2} (y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta \\ \delta |y - \hat{y}| - \frac{1}{2} \delta^2 & \text{otherwise} \end{cases}$$

If the data point is not an outlier

↑
works as MAE if data point is outlier

δ is a hyperparameter.

- Huber loss is used in regression when the dataset may contain outliers.
- It combines the benefits of MAE & MSE.
- It behaves like MSE for small errors & like MAE for large errors, so it is less sensitive to outliers than MSE.

② Binary cross Entropy (log-loss)

Loss function $= -y \log(\hat{y}) - (1-y) \log(1-\hat{y})$

Cost function $= -\frac{1}{n} \sum_{i=1}^n y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)$

- Binary cross entropy is used for classification problems & there must be only two classes like (yes/no, 0/1)
- To use this loss function, the activation function must be sigmoid of output neuron.

Advantages

* Differentiable

disadvantages

* multi local minima

* Not intuitive

③ categorical cross entropy [used in Softmax Regression]

- Used in classification scenarios
- for multi-class classification

$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j)$$

- When we have 3 or more categories in output then used

K - total no. of classes.

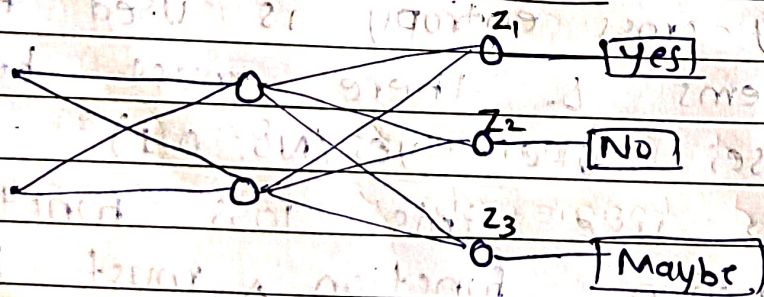
ex. cpga is placed?

	yes	No	maybe
80	yes	0	0
60	No	0	1
70	maybe	0	0

where $K=3$

$$L = -y_1 \log(\hat{y}_1) - y_2 \log(\hat{y}_2) - y_3 \log(\hat{y}_3)$$

* used The activation function of output layer must be Softmax



for No,

$$f(z)_{\text{softmax}} = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

for maybe,

$$f(z) = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$\text{cost function} = -\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^k y_{ij} \log(\hat{y}_{ij})$$

④ sparse categorical cross entropy

- same as categorical cross entropy but.
- * It do not require one hot encoding
- * It uses integer class numbers.

ex., cgpa is placed

7	70	yes	1	we do
8	80	No	2	integer
6	60	maybe	3	encoding

- It is faster
- we only calculate the required log

ex., for output 2 we solve only $-y_2 \log(\hat{y}_2)$